

Alec Lavoie

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Software generalist, driven by ambitious teams taking on problems that feel impossible. I pursue problems worth solving across product vision, user experience, and systems with ruthless agency. Seeking projects that justify obsession.

Experience

Istari Digital

Remote

SENIOR DESIGN ENGINEER • AEROSPACE INTEGRATION & SIMULATION PLATFORM

Apr 2025 – Jun 2026

- Modernized the enterprise product experience by migrating roughly 75% of the application to a new design system—branding, chrome, layout primitives, and dark mode—using a Strangler-pattern deprecation of the legacy system
- Solved cross-subcontractor collaboration in large defense programs by enabling teams to decompose complex projects into versioned primitives shared across permission boundaries, via composable subsystems, an IDE-style workspace, and system-scoped documents, shipping for the AIAA AVIATION Forum 2026 demonstration
- Delivered platform performance improvements for the Connecticut Aerospace & Defense Conference (CADC) 2025 and beyond by halving client request volume via pagination and lazy hydration, then roughly 3x'ing warm-cache responses on a major list endpoint (700ms to 220ms) via permission caching
- Accelerated development velocity by reducing CI runtime from ~15 to 4–6 minutes through parallelized tests and merged coverage
- Stabilized client performance under connection and read pressure constraints with an architecture-specific custom sync implementation that incrementally patches views from the Registry service, avoiding expensive list requests and permissioning database load
- Secured enterprise adoption with a major defense contractor via a Confluence integration for viewing secure platform resources in customer documentation hubs

Revision Labs

Boston, MA

FOUNDER • INDEPENDENT AI PRODUCT EXPERIMENTS — WIKIWIZARD, CINEGIST, REVISION

Jan 2022 – Apr 2025

Dispense

Remote

SENIOR SOFTWARE ENGINEER • E-COMMERCE STOREFRONT BUILDER

July 2022 – Feb 2023

- Empowered customers to dynamically manage their product catalog pricing by engineering a customizable discount and promotion workflow
- Securely managed and distributed keys and secrets with a fingerprint-based CLI tool
- Addressed degrading NoSQL data health by conducting an in-depth data integrity audit using schema introspection tools, and architected a forward-looking reconciliation strategy

Indigo Agriculture

Remote

FRONT-END SOFTWARE ENGINEER • TRANSPORT MANAGEMENT SYSTEM

July 2020 – Jan 2022

- Rolled out fully featured multi-sided logistics marketplace for agriculture product transportation
- Documented and resolved page-by-page application lighthouse warnings, critical workflow frame jank, gql cache misses, and slashed dependency graph bloat by 60%

Appcues

Boston, MA

SOFTWARE ENGINEER • FLOW BUILDER CHROME EXTENSION

July 2019 – April 2020

- Significantly improved application boot time and reliability by orchestrating a seamless migration from Firebase to PostgreSQL using a dual-read-write shim API

General Indemnity Group

Boston, MA

SOFTWARE ENGINEER • SELF-SERVICE SURETY

May 2018 – July 2019

- Unveiled a mobile-optimized, high-performance e-commerce storefront to feature a comprehensive bond product catalog
- Slashed operational costs by 40% and bolstered content scalability through the design and integration of a headless CMS and a GUI admin portal for the e-commerce site

AspenTech

Bedford, MA

SOFTWARE DEVELOPER • UNIVERSAL PROCESS INDUSTRY MANAGEMENT SYSTEM

Nov 2016 – Apr 2018

Skills

Typescript, React, Node.js, SPA, Tailwind, Figma, Cursor, PostgreSQL, GraphQL, ElectricSQL, SpiceDB, authentication & identity, permissioning, caching & performance, event logs & processing, Canvas, XState, Elixir, CI/CD, accessibility, product vision & delivery

Education

Jan 2024 **Machine Learning: Regression and Classification**, DeepLearning.AI, Stanford University

Remote

Jun 2016 **BSc. Computer Science**, University of Massachusetts Dartmouth

Dartmouth, MA